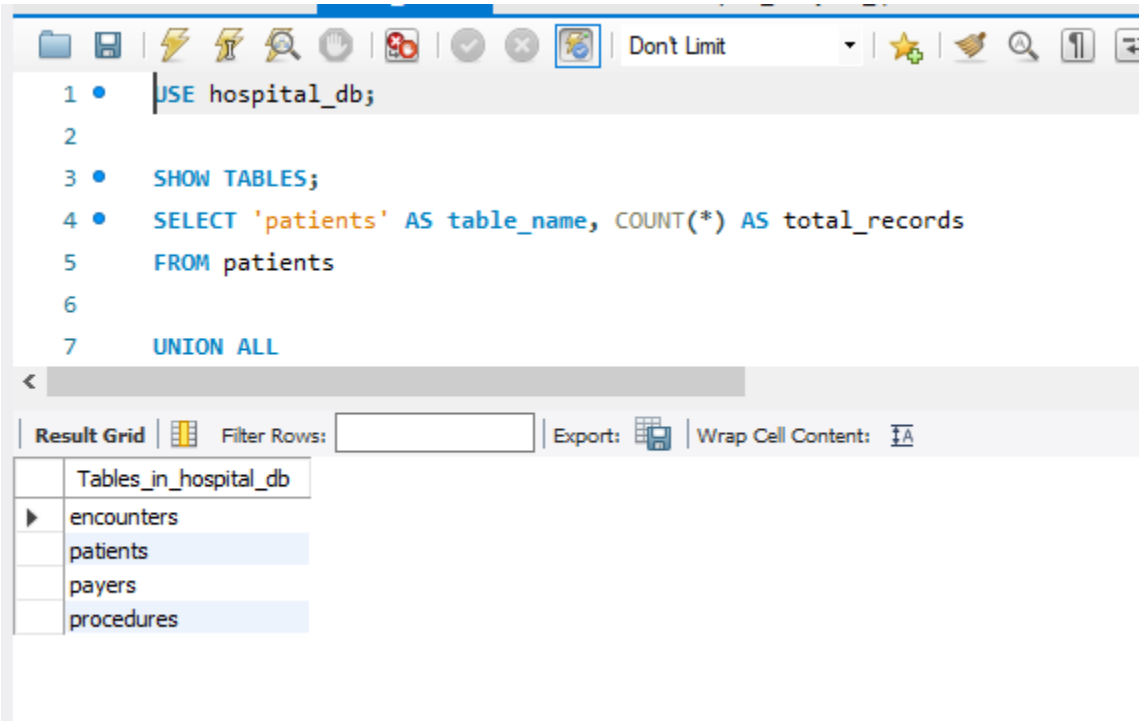
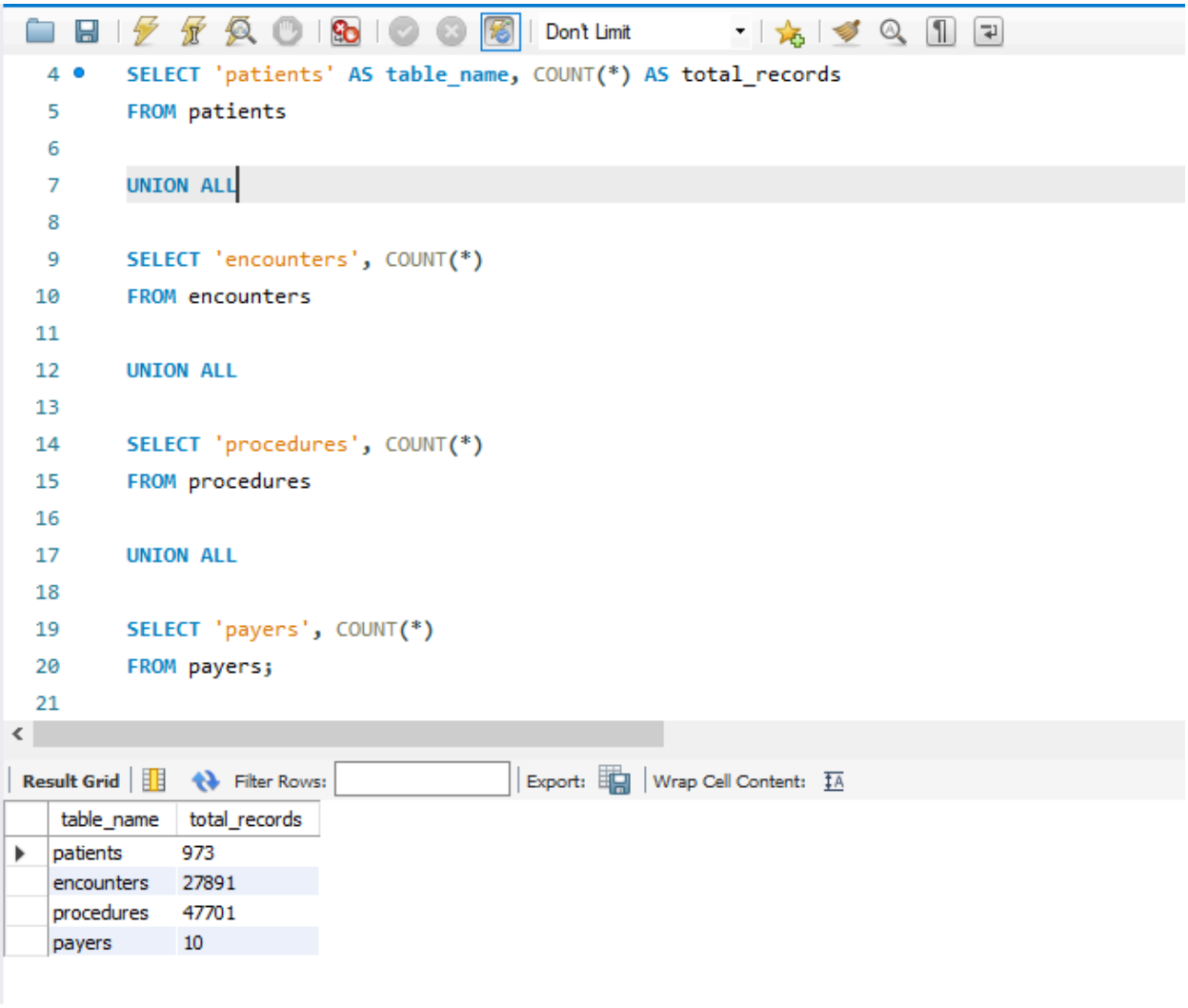
Task 2: Inspect the Datasets

Files and Tables Used:

- `encounters.csv` (27,891 records, 14 columns)
- `patients.csv` (973 records)
- `procedures.csv` (procedure execution records, base costs)
- `payers.csv` (10 records detailing insurance providers and coverage structures)
-





```
SELECT 'patients' AS table_name, COUNT(*) AS total_records
FROM patients

UNION ALL

SELECT 'encounters', COUNT(*)
FROM encounters

UNION ALL

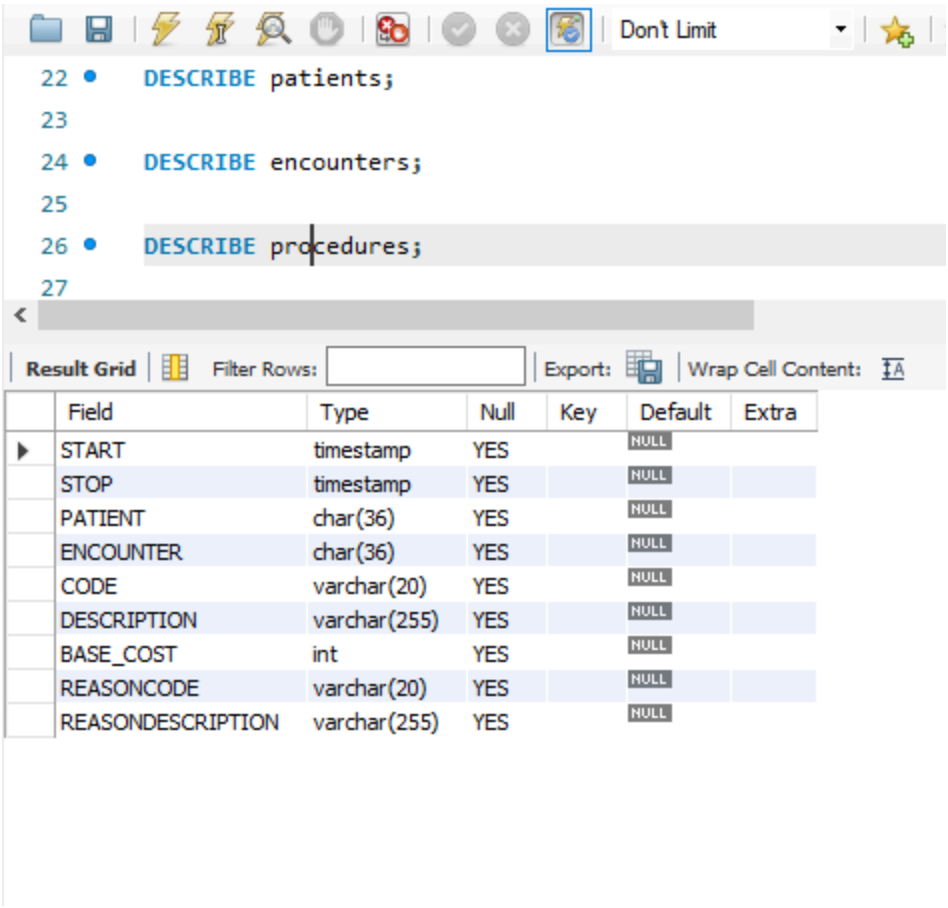
SELECT 'procedures', COUNT(*)
FROM procedures

UNION ALL

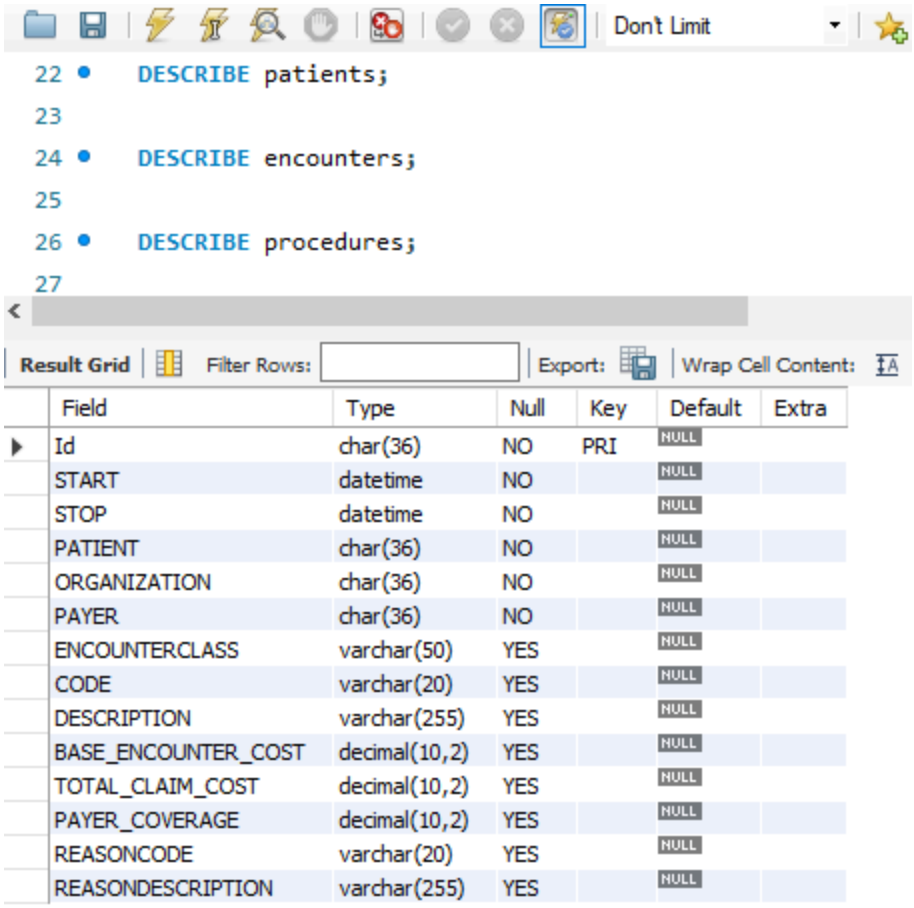
SELECT 'payers', COUNT(*)
FROM payers;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

table_name	total_records
patients	973
encounters	27891
procedures	47701
payers	10



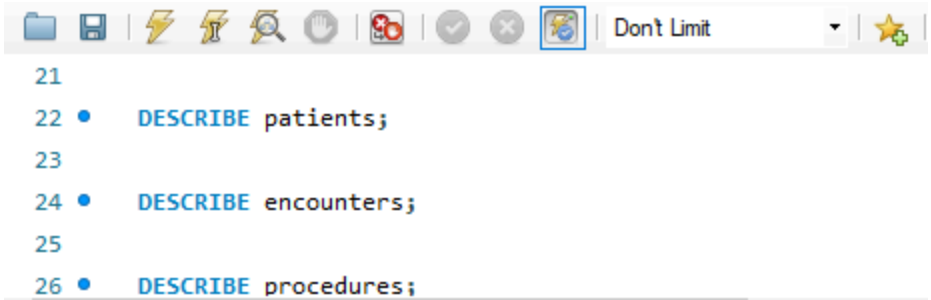
Field	Type	Null	Key	Default	Extra
START	timestamp	YES		NULL	
STOP	timestamp	YES		NULL	
PATIENT	char(36)	YES		NULL	
ENCOUNTER	char(36)	YES		NULL	
CODE	varchar(20)	YES		NULL	
DESCRIPTION	varchar(255)	YES		NULL	
BASE_COST	int	YES		NULL	
REASONCODE	varchar(20)	YES		NULL	
REASONDESCRIPTION	varchar(255)	YES		NULL	



```
22 •  DESCRIBE patients;
23
24 •  DESCRIBE encounters;
25
26 •  DESCRIBE procedures;
27
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

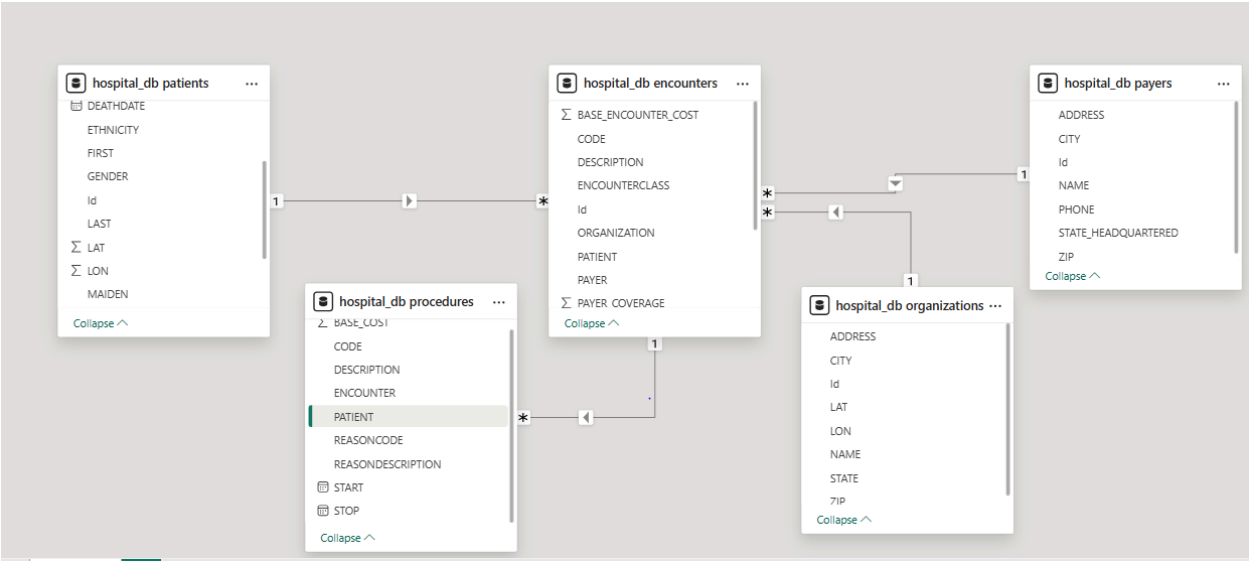
Field	Type	Null	Key	Default	Extra
Id	char(36)	NO	PRI	NULL	
START	datetime	NO		NULL	
STOP	datetime	NO		NULL	
PATIENT	char(36)	NO		NULL	
ORGANIZATION	char(36)	NO		NULL	
PAYER	char(36)	NO		NULL	
ENCOUNTERCLASS	varchar(50)	YES		NULL	
CODE	varchar(20)	YES		NULL	
DESCRIPTION	varchar(255)	YES		NULL	
BASE_ENCOUNTER_COST	decimal(10,2)	YES		NULL	
TOTAL_CLAIM_COST	decimal(10,2)	YES		NULL	
PAYER_COVERAGE	decimal(10,2)	YES		NULL	
REASONCODE	varchar(20)	YES		NULL	
REASONDESCRIPTION	varchar(255)	YES		NULL	



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Id	char(36)	NO	PRI	NULL	
BIRTHDATE	date	YES		NULL	
DEATHDATE	date	YES		NULL	
PREFIX	varchar(10)	YES		NULL	
FIRST	varchar(100)	YES		NULL	
LAST	varchar(100)	YES		NULL	
SUFFIX	varchar(10)	YES		NULL	
MAIDEN	varchar(100)	YES		NULL	
MARITAL	char(1)	YES		NULL	
RACE	varchar(50)	YES		NULL	
ETHNICITY	varchar(50)	YES		NULL	
GENDER	char(1)	YES		NULL	
BIRTHPLACE	varchar(255)	YES		NULL	
ADDRESS	varchar(255)	YES		NULL	
CITY	varchar(100)	YES		NULL	
STATE	varchar(100)	YES		NULL	
COUNTY	varchar(100)	YES		NULL	
ZIP	varchar(10)	YES		NULL	
LAT	double	YES		NULL	
LON	double	YES		NULL	

Relationships
Linked via unique keys



Task 3: Define Success Criteria

- Analytical Reconciliation: Dashboard metrics reconcile 100% with independent SQL queries on total encounters, unique patient counts, and zero-coverage ratios.
- Model Performance: Relationships and filters in Power BI execute cross-filtering smoothly without ambiguity or circular dependencies.
- User Usability: The report answers all core operational questions (costs, procedures, readmissions) within 2 clicks from the overview page.